(notes) 5.2 Polynomials and Power Functions, part 1

Key Terms:

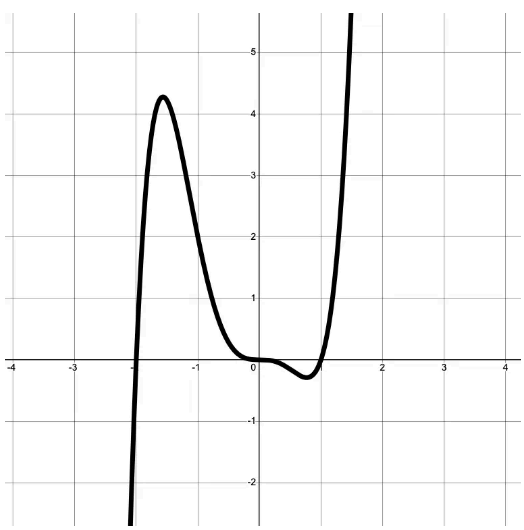
- End Behavior
- Even and Odd functions
- Degree of a Polynomial
- Multiplicity
- Leading Coefficient
- Turning Point

End Behavior

We want to describe the value of the function $f(x)$ for extreme values of x, when $x \to +\infty$ (x approaches positive infinity) and when $x \to -\infty$ (x approaches negative infinity).

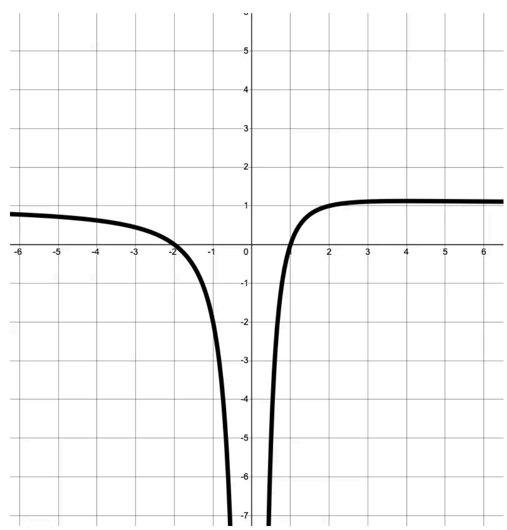
In other words, what are the y values when we look to the extreme right (x approaches positive infinity) and what are the y values when we look to the extreme left (x approaches negative infinity).

Example



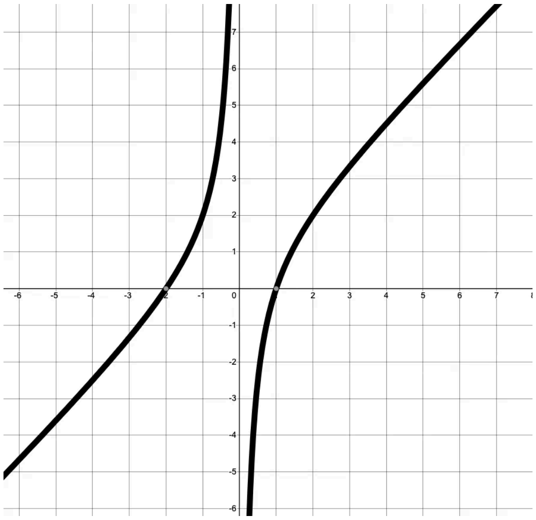
when $x \to +\infty$, $f(x) \to +\infty$
"as x approaches positive infinity to the right, the function goes off to infinity"

when $x \to -\infty$, $f(x) \to -\infty$
"as x approaches negative infinity to the left, the function goes off to negative infinity"



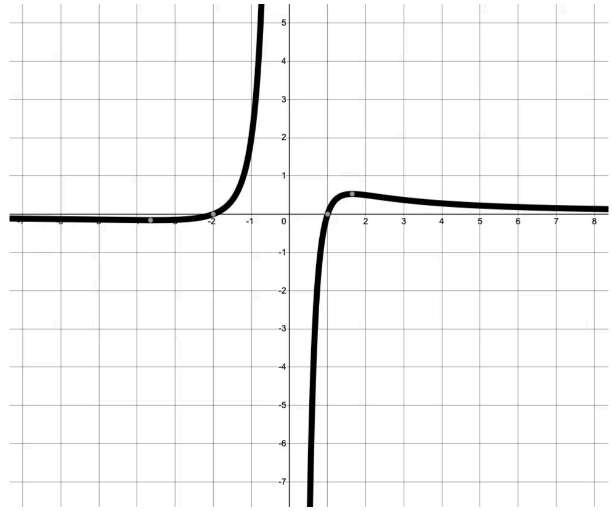
when $x \to +\infty$, $f(x) \to 1$
"as x approaches positive infinity to the right, the function approaches a value of 1"

when $x \to -\infty$, $f(x) \to 1$
"as x approaches negative infinity to the left, the function approaches a value of 1"



when $x \to +\infty$, $f(x) \to$ _____
"as x approaches positive infinity to the right, the function approaches _____"

when $x \to -\infty$, $f(x) \to$ _____
"as x approaches negative infinity to the left, the function approaches _____"

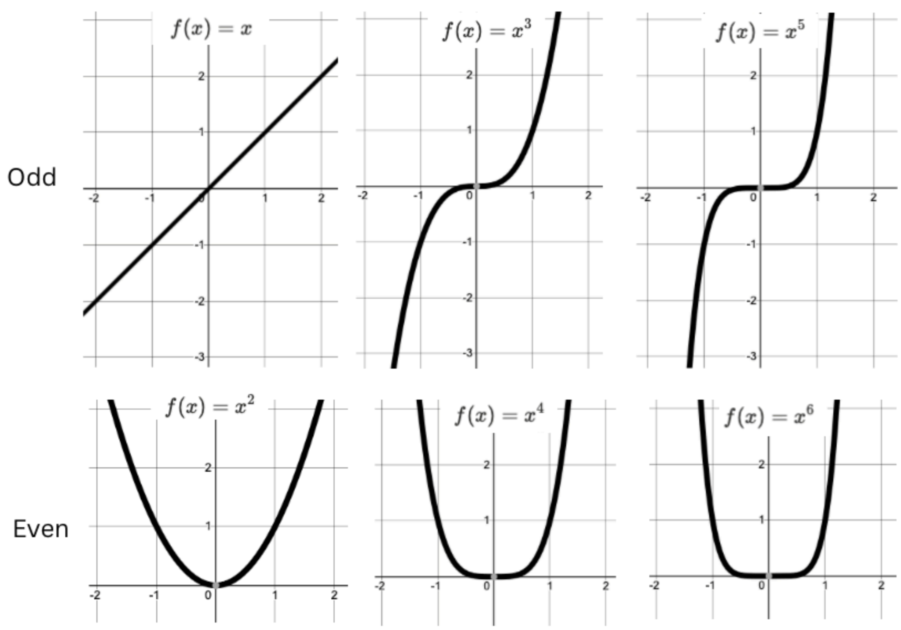


when $x \to +\infty$, $f(x) \to$ _____
"as x approaches positive infinity to the right, the function approaches _____"

when $x \to -\infty$, $f(x) \to$ _____
"as x approaches negative infinity to the left, the function approaches _____"

Even and Odd Functions

The even/odd refers to the highest degree/power on the polynomial. That highest degree will determine the overall shape of the function.

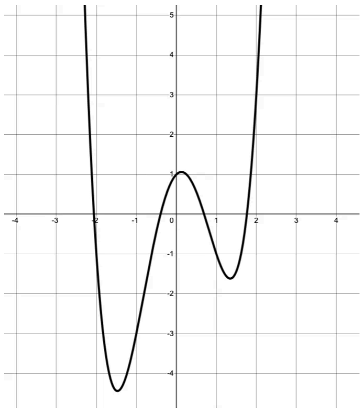


Discuss, even/odd functions:

How can you generalize the end behavior for **even** functions and **odd** functions?

Diagramming Polynomials

$f(x) = x^4 - 4x^2 + x + 1$



Degree of polynomial: ______

Even or Odd: ______

Leading Coefficient: ______

Number of intercepts: ______

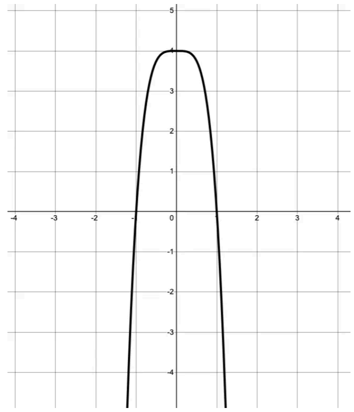
Number of min/maxes: ______

End behavior:

as $x \to -\infty$, $f(x) =$

as $x \to +\infty$, $f(x) =$

$f(x) = -4x^4 + 4$



Degree of polynomial: ______

Even or Odd: ______

Leading Coefficient: ______

Number of intercepts: ______

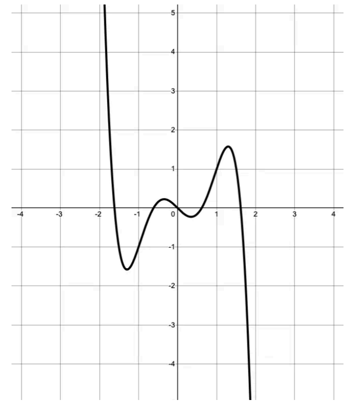
Number of min/maxes: ______

End behavior:

as $x \to -\infty$, $f(x) =$

as $x \to +\infty$, $f(x) =$

$f(x) = -x^5 + 3x^3 - x$



Degree of polynomial: ______

Even or Odd: ______

Leading Coefficient: ______

Number of intercepts: ______

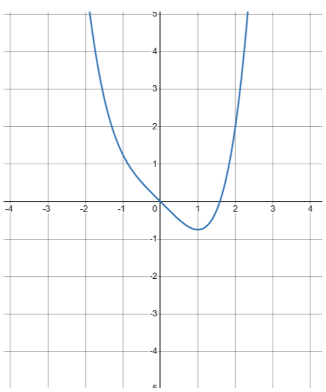
Number of min/maxes: ______

End behavior:

as $x \to -\infty$, $f(x) =$

as $x \to +\infty$, $f(x) =$

$f(x) = \frac{1}{4}x^4 - x$



Degree of polynomial: ______

Even or Odd: ______

Leading Coefficient: ______

Number of intercepts: ______

Number of min/maxes: ______

End behavior:

as $x \to -\infty$, $f(x) =$

as $x \to +\infty$, $f(x) =$

Discuss

1. How is the **degree** of the polynomial related to the **number of intercepts**? number of mins/maxes?
2. How does the **even/oddness** of a function and the **leading coefficient** influence the **end behavior**?

x-intercepts

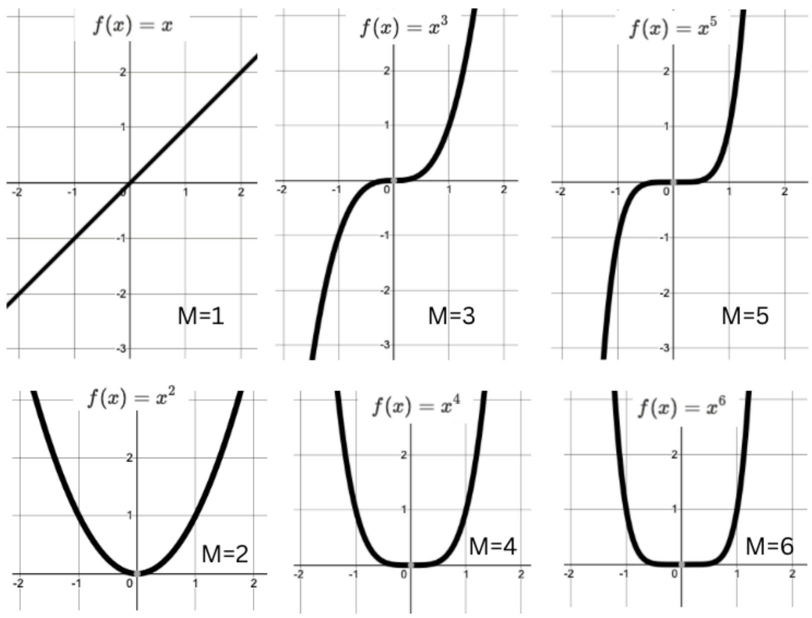
f a polynomial is in factored form, you can pull out the x-intercepts very easily! Recall x-intercepts nay be referred to as “zeroes” or “roots” of a function.

Multiplicity, M

The **multiplicity, M,** of the root is the exponent on each unique factor. The multiplicity tells you how the graph crosses the x axis at that intersection point.

Cross x axis

Bounce on x axis



Example: $f(x) = x^3(x+2)^2(x-1)$

There are 3 unique factors and so there are 3 intercepts at $x = -2, 0, 1$.

x-intercept: -2

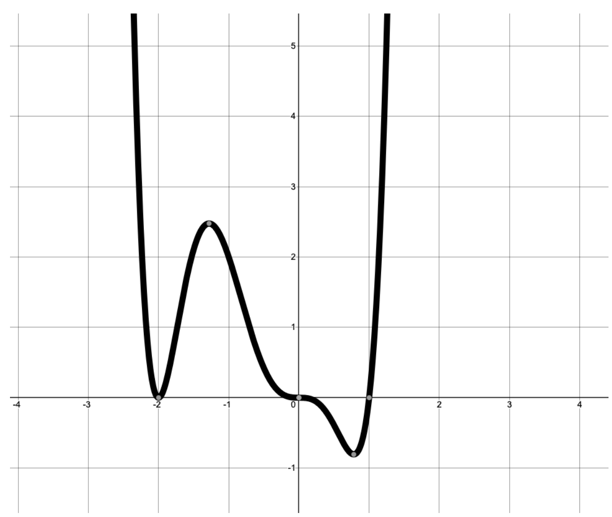
The $(x+2)^2$ root is squared and has multiplicity $M = 2$ and so intersects the x-axis like a quadratic function at $x = -2$.

x-intercept: 0

The x^3 root is squared and has multiplicity $M = 3$ and so intersects the x-axis like a cubic function at $x = 0$.

x-intercept: 1

The $(x-1)$ root is squared and has multiplicity $M = 1$ and so intersects the x-axis linearly at $x = 1$.



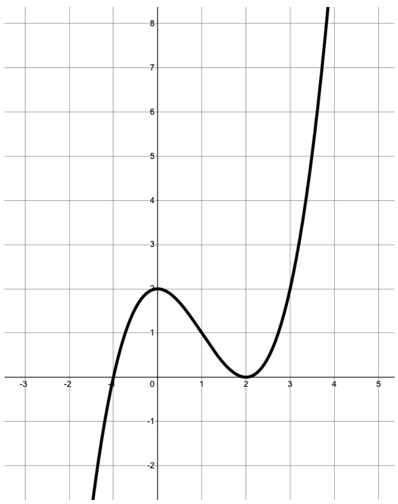
Three intercepts at x = -2, 0, 1 with different multiplicities.

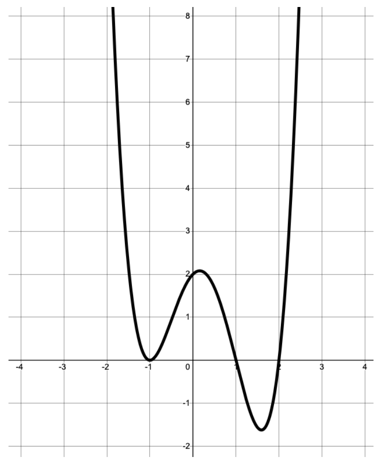
Check In: What is the degree of this polynomial? Is it even or odd? What is the end behavior for $x \to \pm\infty$?

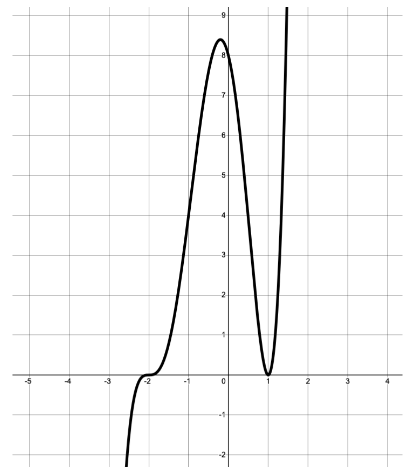
Practice

Write a possible equation for the following plots. Find the x-intercepts to write the factors. Then find the multiplicity for each root to give each factor the correct power.

Once you have done these, go back to the diagramming section and write possible equations for each plot 🙂







Example:

x-intercepts: $x = -1, 2$

At $x = -1$, the graph crosses linearly. Thus, $M = 1$ and the factor is: $(x+1)^1$

At $x = 2$, the graph touches quadratically. Thus, $M = 2$ and the factor is: $(x-2)^2$

A possible equation:

$f(x) = (x+1)(x-2)^2$

x-intercepts:

At $x =$

$M =$

The factor is:

At $x =$

$M =$

The factor is:

At $x =$

$M =$

The factor is:

End behavior:
Leading Coefficient: + / -

A possible equation:

x-intercepts:

At $x =$

$M =$

The factor is:

At $x =$

$M =$

The factor is:

End behavior:
Leading Coefficient: + / -

A possible equation: